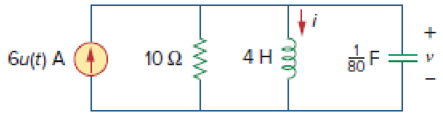
Chapter 16, Problem 63P

Problem

Consider the parallel *RLC* circuit of Fig. 16.86. Find $v(t)$ and $i(t)$ given that $v(0) = 7.5$ V and $i(0) = -3$ A.

Figure 16.86:



Step-by-step solution

Step 1 of 6

Refer to the Figure 16.86 in the textbook.

Redraw the circuit in s-domain as shown in Figure 1.

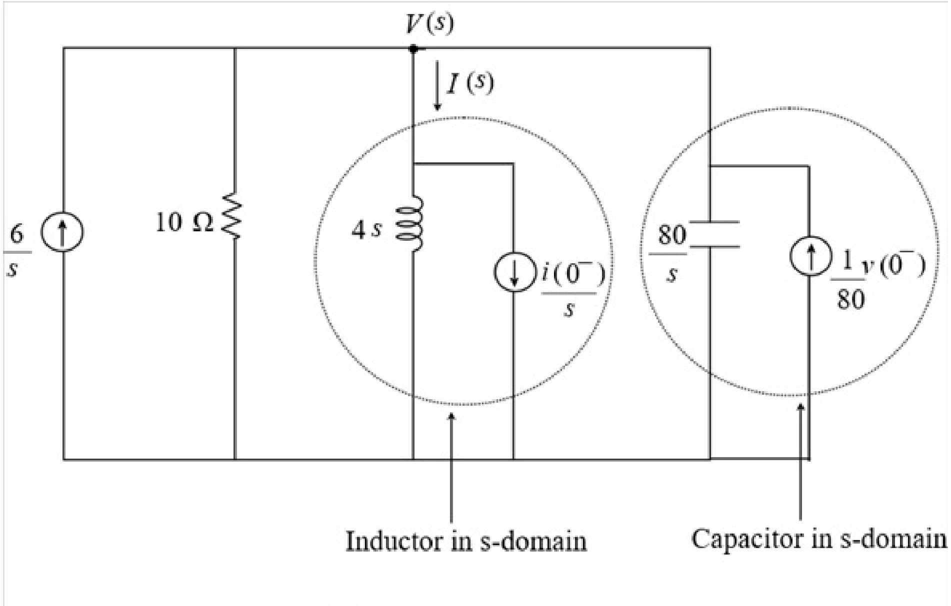


Figure 1

Step 2 of 6

Apply Kirchhoff's current law in Figure 1.

$$I(s)+\frac{V(s)}{10}-\frac{6}{s}+\frac{V(s)}{\left(\frac{80}{s}\right)}-\frac{v(0^-)}{80}=0$$

$$\frac{V(s)}{4s}+\frac{i(0^-)}{s}+\frac{V(s)}{10}-\frac{6}{s}+\frac{sV(s)}{80}-\frac{v(0^-)}{80}=0$$

Substitute $7.5\,\text{V}$ for $v(0^-)$ and $-3\,A$ for $i(0^-)$.

$$\frac{V(s)}{4s}-\frac{3}{s}+\frac{V(s)}{10}-\frac{6}{s}+\frac{sV(s)}{80}-\frac{7.5}{80}=0$$

$$\frac{V(s)}{4s}+\frac{V(s)}{10}+\frac{sV(s)}{80}=\frac{9}{s}+0.093$$

$$\frac{(20+8s+s^2)V(s)}{80s}=\frac{9+0.093s}{s}$$

$$V(s)=\frac{720+7.44s}{s^2+8s+20}$$

Step 3 of 6

Take the inverse Laplace transform of $V(s)$ to obtain $v(t)$.

$$v(t)=\text{L}^{-1}\left[\frac{7.44s+720}{(s^2+8s+20)}\right]$$

$$=7.44\text{L}^{-1}\left[\frac{s}{s^2+8s+20}\right]+\text{L}^{-1}\left[\frac{720}{s^2+8s+20}\right]$$

$$=7.44\text{L}^{-1}\left[\frac{s}{s^2+(2)(4)(1)s+4^2-4^2+20}\right]+\text{L}^{-1}\left[\frac{720}{s^2+(2)(4)(1)s+4^2-4^2+20}\right]$$

$$=7.44\text{L}^{-1}\left[\frac{s}{(s+4)^2+2^2}\right]+\text{L}^{-1}\left[\frac{720}{(s+4)^2+2^2}\right]$$

$$=7.44\text{L}^{-1}\left[\frac{s+4}{(s+4)^2+2^2}\right]+\text{L}^{-1}\left[\frac{-(7.44)(4)+720}{(s+4)^2+2^2}\right]$$

$$=7.44e^{-4t}\cos(2t)u(t)+\text{L}^{-1}\left[\frac{690.24}{(s+4)^2+2^2}\right]$$

$$=7.44e^{-4t}\cos(2t)u(t)+\frac{690.24}{2}\text{L}^{-1}\left[\frac{2}{(s+4)^2+2^2}\right]$$

$$=\left[7.44e^{-4t}\cos(2t)u(t)+345.12\sin(2t)u(t)\right]\text{V}$$

Hence, the value of $v(t)$ is $\boxed{\left[7.44e^{-4t}\cos(2t)+345.12e^{-4t}\sin(2t)\right]u(t)\ \text{V}}$.

Comments (1)

Anonymous

Not the same as the back of the book.

Step 4 of 6

Write the expression of $I(s)$.

$$I(s)=\frac{V(s)}{4s}+\frac{i(0^-)}{s}$$

Substitute $\frac{720+7.44s}{s^2+8s+20}$ for $V(s)$ and $-3\,A$ for $i(0^-)$.

$$\begin{aligned}I(s)&=\frac{\left(\frac{720+7.44s}{s^2+8s+20}\right)}{4s}-\frac{3}{s}\\&=\frac{1}{4s}\left(\frac{720+7.44s}{s^2+8s+20}\right)-\frac{3}{s}\end{aligned}$$

Step 5 of 6

Convert $\frac{1}{4s}\left(\frac{720+7.44s}{s^2+8s+20}\right)$ into its partial fraction form.

$$\begin{aligned}\frac{1.86s+180}{s\left(s^2+8s+20\right)}&=\frac{A}{s}+\frac{Bs+C}{s^2+8s+20}\\&=\frac{A\left(s^2+8s+20\right)+\left(Bs+C\right)s}{s\left(s^2+8s+20\right)}\end{aligned}$$

Compare the coefficients of s^2, s and the constant terms on both the sides.

$A+B=0$ …… (1)

$8A+C=1.86$ …… (2)

$20A=180$

$A=9$

Substitute 9 for A in equation (1) and (2).

$B=-9$

$C=-70.14$

Write the partial fraction form of $\frac{1}{4s}\left(\frac{720+7.44s}{s^2+8s+20}\right)$.

$$\frac{1}{4s}\left(\frac{720+7.44s}{s^2+8s+20}\right)=\frac{9}{s}+\frac{-9s-70.14}{s^2+8s+20}$$

Substitute $\frac{9}{s}+\frac{-9s-70.14}{s^2+8s+20}$ for $\frac{1}{4s}\left(\frac{720+7.44s}{s^2+8s+20}\right)$ in the expression of $I(s)$.

$$\begin{aligned}I(s)&=\frac{9}{s}+\frac{-9s-70.14}{s^2+8s+20}-\frac{3}{s}\\&=\frac{6}{s}-\frac{9s}{s^2+8s+20}-\frac{70.14}{s^2+8s+20}\end{aligned}$$

Step 6 of 6

Take the inverse Laplace transform of $I(s)$ to obtain $i(t)$.

$$\begin{aligned}i(t)&=\mathrm{L}^{-1}\left[\frac{6}{s}\right]-9\mathrm{L}^{-1}\left[\frac{s+4}{(s+4)^2+2^2}\right]-\mathrm{L}^{-1}\left[\frac{70.14-36}{(s+4)^2+2^2}\right]\\&=6u(t)-9e^{-4t}\cos(2t)u(t)-\frac{34.14}{2}\mathrm{L}^{-1}\left[\frac{2}{(s+4)^2+2^2}\right]\\&=\left[6-9e^{-4t}\cos(2t)-17.07e^{-4t}\sin(2t)\right]u(t)\text{ A}\end{aligned}$$

Hence, the value of $i(t)$ is $\boxed{\left[6-9e^{-4t}\cos(2t)-17.07e^{-4t}\sin(2t)\right]u(t)\text{ A}}$.